Definition 24 (Absolutely Convergent Series)**.** *An infinite series*

$$\sum_{n=1}^{\infty} a_n$$

is said to be absolutely convergent *if the series*

$$\sum_{n=1}^{\infty} |a_n|$$

is convergent.

Remark 6. *If $a_n \geq 0$ for every n, then*

$$|a_n| = a_n,$$

and hence

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} a_n.$$

Therefore, a series of non-negative terms is convergent if and only if it is absolutely convergent.

Definition 25 (Conditionally Convergent Series)**.** *A series is said to be* conditionally convergent *if it is convergent but not absolutely convergent.*

Example 41. *Consider the constant sequence*

$$a_n = c, \qquad n \in \mathbb{N},$$

where $c \in \mathbb{R}$ is fixed. Determine whether the series

$$\sum_{n=1}^{\infty} a_n$$

is convergent.
Solution. The nth partial sum is

$$s_n = a_1 + \cdots + a_n = nc.$$

If $c = 0$, then $s_n = 0$ for every n, and hence the series converges.
Now suppose $c \neq 0$. By the Archimedean Property,

$$|nc| \longrightarrow \infty.$$

Therefore, the sequence (s_n) is divergent, and hence the series is divergent. Thus,

$$\sum_{n=1}^{\infty} a_n$$

is convergent if and only if $c = 0$.

Example 42 (Geometric Series)**.** *Let $z \in \mathbb{R}$ satisfy $|z| < 1$. Consider the infinite series*

$$\sum_{n=0}^{\infty} z^n.$$

Since the nth partial sum is

$$s_n = \sum_{k=0}^{n} z^k = \frac{1 - z^{n+1}}{1 - z},$$

and

$$z^{n+1} \longrightarrow 0,$$

it follows that

$$s_n \longrightarrow \frac{1}{1-z}.$$

Hence,

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}, \qquad |z| < 1.$$

Example 43 (Telescoping Series)**.** *Let (a_n) and (b_n) be two sequences satisfying*

$$a_n = b_{n+1} - b_n, \qquad n \in \mathbb{N}.$$

Then the nth partial sum is

$$\begin{aligned} s_n &= a_1 + a_2 + \cdots + a_n \\ &= (b_2 - b_1) + (b_3 - b_2) + \cdots + (b_{n+1} - b_n) \\ &= b_{n+1} - b_1. \end{aligned}$$

Hence, the series

$$\sum_{n=1}^{\infty} a_n$$

is convergent if and only if the sequence (b_n) is convergent. In that case,

$$\sum_{n=1}^{\infty} a_n = -b_1 + \lim_{n\to\infty} b_n.$$

Example 44. *Evaluate the series*

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

Solution. Observe that

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Thus,

$$a_n = \frac{1}{n(n+1)}, \qquad b_n = -\frac{1}{n}.$$

Since

$$\lim_{n\to\infty} b_n = 0,$$

the previous example gives

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = -b_1 + \lim_{n\to\infty} b_n = 1.$$

Example 45 (Harmonic Series)**.** *Recall Example 18. The harmonic series*

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

is divergent.

Example 46 (p-Series)**.** *For $p > 0$, consider the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. Then*

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

is convergent if and only if $p > 1$. Equivalently,

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \begin{cases} \textit{convergent}, & p > 1, \\ \textit{divergent}, & 0 < p \leq 1. \end{cases}$$

In particular, the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent, whereas the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

Theorem 18 (Cauchy Criterion)**.** *The series $\sum_{n=1}^{\infty} a_n$ converges if and only if for* *every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that*

$$n, m \geq N \quad \Longrightarrow \quad |s_n - s_m| < \varepsilon,$$

where (s_n) denotes the sequence of partial sums.
Thus, the series $\sum_{n=1}^{\infty} a_n$ converges if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n > m \geq N \quad \Longrightarrow \quad |a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon.$$

Proof. The series $\sum_{n=1}^{\infty} a_n$ converges if and only if the sequence (s_n) of partial sums converges. By the Cauchy Criterion for sequences, this is equivalent to saying that (s_n) is a Cauchy sequence. Since

$$s_n - s_m = a_{m+1} + a_{m+2} + \cdots + a_n,$$

the second statement follows immediately. □

Theorem 19. *If the series $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \to 0$.*

Proof. Let $s_n = \sum_{k=1}^{n} a_k$ be the sequence of partial sums. Since the series converges, (s_n) converges. Hence, by the Cauchy Criterion for sequences, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|s_n - s_m| < \varepsilon, \qquad \forall\, n, m \geq N.$$

Now let $n \geq N + 1$. Then $n - 1 \geq N$, and therefore $|a_n| = |s_n - s_{n-1}| < \varepsilon$. Hence $a_n \to 0$. □

Theorem 20. *Suppose that the series $\sum_{n=1}^{\infty} a_n$ converges to s. Then, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$\left| \sum_{n=N+1}^{\infty} a_n \right| < \varepsilon.$$

Proof. Let $\varepsilon > 0$ be given. Since $s_n \to s$, there exists $N \in \mathbb{N}$ such that $|s - s_N| < \varepsilon$. Since $\sum_{n=N+1}^{\infty} a_n = s - s_N$, it follows that

$$\left| \sum_{n=N+1}^{\infty} a_n \right| = |s - s_N| < \varepsilon.$$

This completes the proof.
Note: For some $k \in \mathbb{N}$, the sum $\sum_{n=k}^{\infty} a_n$ is called the tail of the series. □

Theorem 21 (Algebra of Convergent Series)**.** *Let* $\sum_{n=1}^{\infty} a_n$ *and* $\sum_{n=1}^{\infty} b_n$ *be two convergent series with sums* A *and* B*, respectively. Then:*

1. *The series* $\sum_{n=1}^{\infty}(a_n + b_n)$ *is convergent and*

$$\sum_{n=1}^{\infty}(a_n + b_n) = A + B.$$

2. *For every* $\lambda \in \mathbb{R}$*, the series* $\sum_{n=1}^{\infty} \lambda a_n$ *is convergent and*

$$\sum_{n=1}^{\infty} \lambda a_n = \lambda A.$$

Theorem 22. *If the series* $\sum_{n=1}^{\infty} |a_n|$ *converges, then the series* $\sum_{n=1}^{\infty} a_n$ *also converges.*

Proof. Let $s_n = \sum_{k=1}^{n} a_k$ and $\sigma_n = \sum_{k=1}^{n} |a_k|$ denote the sequences of partial sums of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} |a_n|$, respectively.
Since (σ_n) converges, it is a Cauchy sequence. Thus, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|\sigma_n - \sigma_m| < \varepsilon, \qquad \forall\, n > m \geq N.$$

For $n > m \geq N$, we have

$$|s_n - s_m| = \left| \sum_{k=m+1}^{n} a_k \right| \leq \sum_{k=m+1}^{n} |a_k| = \sigma_n - \sigma_m < \varepsilon.$$

Hence, (s_n) is a Cauchy sequence. By the Cauchy Criterion for series, the series $\sum_{n=1}^{\infty} a_n$ converges. □

3.2 Test of convergence

Theorem 23 (Comparison Test)**.** *Let* $\sum_{n=1}^{\infty} a_n$ *and* $\sum_{n=1}^{\infty} b_n$ *be two series of non-negative real numbers. Suppose that* $a_n \leq b_n$ *for all* $n \in \mathbb{N}$*. Then:*

1. *If* $\sum_{n=1}^{\infty} b_n$ *converges, then* $\sum_{n=1}^{\infty} a_n$ *also converges.*
2. *If* $\sum_{n=1}^{\infty} a_n$ *diverges, then* $\sum_{n=1}^{\infty} b_n$ *also diverges.*

Example 47. *The series* $\sum_{n=1}^{\infty} \frac{1}{n^2}$ *is convergent. Since*

$$0 \le \frac{1}{n^3} \le \frac{1}{n^2}, \qquad \forall\, n \in \mathbb{N},$$

the Comparison Test implies that $\sum_{n=1}^{\infty} \frac{1}{n^3}$ *is also convergent.*

Example 48. *The harmonic series* $\sum_{n=1}^{\infty} \frac{1}{n}$ *is divergent. Since*

$$\frac{1}{n} \le \frac{1}{\sqrt{n}}, \qquad \forall\, n \in \mathbb{N},$$

the Comparison Test implies that $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ *is also divergent.*

Theorem 24 (d'Alembert's Ratio Test)**.** *Let* $\sum_{n=1}^{\infty} a_n$ *be a series of positive real numbers. Suppose that*

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = r.$$

Then:

1. *If* $0 \le r < 1$*, then the series* $\sum_{n=1}^{\infty} a_n$ *converges.*
2. *If* $r > 1$*, then the series* $\sum_{n=1}^{\infty} a_n$ *diverges.*
3. *If* $r = 1$*, then the test is inconclusive.*

Example 49. *Consider the series* $\sum_{n=1}^{\infty} \dfrac{1}{n!}$*. Since*

$$\frac{a_{n+1}}{a_n} = \frac{1}{n+1} \longrightarrow 0,$$

the Ratio Test implies that the series converges.

Example 50. *Consider the series* $\sum_{n=1}^{\infty} \dfrac{2^n}{n}$*. Since*

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{n+1} \cdot \frac{n}{2^n} = 2 \cdot \frac{n}{n+1} \longrightarrow 2 > 1,$$

the Ratio Test implies that the series diverges.

Example 51. *Consider the harmonic series* $\sum_{n=1}^{\infty} \dfrac{1}{n}$*. Since*

$$\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \longrightarrow 1,$$

the Ratio Test is inconclusive. Indeed, the harmonic series is divergent.
On the other hand, for the series $\sum_{n=1}^{\infty} \dfrac{1}{n^2}$*,*

$$\frac{a_{n+1}}{a_n} = \left(\frac{n}{n+1}\right)^2 \longrightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

Theorem 25 (Cauchy's Root Test)**.** *Let* $\sum_{n=1}^{\infty} a_n$ *be a series of positive real numbers. Suppose that*

$$\lim_{n\to\infty} \sqrt[n]{a_n} = r.$$

Then:

1. *If* $0 \leq r < 1$*, then the series* $\sum_{n=1}^{\infty} a_n$ *converges.*
2. *If* $r > 1$*, then the series* $\sum_{n=1}^{\infty} a_n$ *diverges.*
3. *If* $r = 1$*, then the test is inconclusive.*

Example 52. *Consider the series* $\sum_{n=1}^{\infty} \frac{1}{2^n}$*. Since*

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{2^n}} = \frac{1}{2},$$

the Root Test implies that the series converges.

Example 53. *Consider the series* $\sum_{n=1}^{\infty} 2^n$*. Since*

$$\sqrt[n]{a_n} = \sqrt[n]{2^n} = 2,$$

the Root Test implies that the series diverges.

Example 54. *Consider the harmonic series* $\sum_{n=1}^{\infty} \frac{1}{n}$. *Since*

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n}} = \frac{1}{n^{1/n}} \longrightarrow 1,$$

the Root Test is inconclusive. Indeed, the harmonic series is divergent.
On the other hand, for the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$,

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n^2}} = \frac{1}{n^{2/n}} \longrightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

Example 55. *Show that the series* $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ *converges.*

Solution. *Let* $a_n = \dfrac{2^n n!}{n^n}$. *Then*

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{2^n n!} = 2\left(\frac{n}{n+1}\right)^n.$$

Since $\left(1+\frac{1}{n}\right)^n \to e$, *we have*

$$\left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1+\frac{1}{n}\right)^n} \longrightarrow \frac{1}{e}.$$

Hence

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \frac{2}{e} < 1.$$

Therefore, by the Ratio Test, the series $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ *converges.*

Example 56. *Evaluate the following limits:*

1. $\displaystyle\lim_{n\to\infty} \frac{(n!)^{1/n}}{n}$.
2. $\displaystyle\lim_{n\to\infty} \frac{n!}{n^n}$.

Brief Solution.

1. *Taking logarithms,* $\log\left(\dfrac{(n!)^{1/n}}{n}\right) = \dfrac{1}{n}\displaystyle\sum_{k=1}^{n} \log\left(\frac{k}{n}\right)$. *This is a Riemann sum for*

$\int_0^1 \log x\,dx = -1$. *Hence*

$$\lim_{n\to\infty} \frac{(n!)^{1/n}}{n} = \frac{1}{e}.$$

2. *We have* $0 \leq \frac{n!}{n^n} = \prod_{k=1}^{n} \frac{k}{n} \leq \left(\frac{1}{2}\right)^{\lfloor n/2 \rfloor}$. *Since the right-hand side tends to* 0*, the Sandwich Theorem gives*

$$\lim_{n\to\infty} \frac{n!}{n^n} = 0.$$

Example 57. *Determine whether the series* $\sum_{n=1}^{\infty} \frac{7^{n+1}}{9^n}$ *is convergent.*

Solution. *Let* $a_n = \frac{7^{n+1}}{9^n}$. *Then*

$$\frac{a_{n+1}}{a_n} = \frac{7^{n+2}}{9^{n+1}} \cdot \frac{9^n}{7^{n+1}} = \frac{7}{9}.$$

Since

$$\frac{7}{9} < 1,$$

the Ratio Test implies that the series $\sum_{n=1}^{\infty} \frac{7^{n+1}}{9^n}$ *converges.*

Theorem 26 (Abel-Pringsheim Criterion)**.** *Let* (a_n) *be a decreasing sequence of non-negative real numbers. If the series* $\sum_{n=1}^{\infty} a_n$ *converges, then* $na_n \to 0$.

Proof. Since $\sum a_n$ is convergent, there exists N such that $|s_n - s_m| < \varepsilon$ for $n, m \geq N$. For $k \geq N$,

$$ka_{2k} \leq a_{k+1} + \cdots + a_{2k} = s_{2k} - s_k.$$

Thus,

$$\lim_{n\to\infty} (2n)a_{2n} = 0.$$

Since $a_{2n+1} \leq a_{2n}$, we have

$$(2n+1)a_{2n+1} \leq (2n+1)a_{2n} \leq (2n)a_{2n} + a_{2n}.$$

Now the series $\sum a_n$ is convergent, so the sequence (a_n) converges to 0, and hence the subsequence (a_{2n}) converges to zero. It follows from above that the sequence $\big((2n+1)a_{2n+1}\big)$ is convergent. Using Exercise 15, we conclude that $na_n \to 0$. $\square$

Consider the series

$$\sum_{n=2}^{\infty} \frac{1}{n \log n}.$$

Although

$$\lim_{n\to\infty} \frac{n}{n \log n} = 0,$$

the series nevertheless diverges. We will see later, using the integral test, why this series diverges.

Example 58. *Show that the series* $\sum_{n=1}^{\infty} \frac{1}{an+b}$, *where* $a, b > 0$, *is divergent.*

Solution. *Let* $a_n = \frac{1}{an+b}$. *Then* (a_n) *is a decreasing sequence of non-negative real numbers. If the series* $\sum_{n=1}^{\infty} a_n$ *were convergent, then by the Abel–Pringsheim Criterion* (26), *we would have* $na_n \to 0$.
However,

$$na_n = \frac{n}{an+b} = \frac{1}{a+\frac{b}{n}} \longrightarrow \frac{1}{a} \neq 0.$$

This contradiction shows that the series $\sum_{n=1}^{\infty} \frac{1}{an+b}$ *is divergent.*

Exercise 17 (Abel's Test)**.** *Assume that* $\sum a_n$ *is convergent and* (b_n) *is either increasing or decreasing, and there exist* $m, M \in \mathbb{R}$ *such that*

$$m \leq b_n \leq M, \qquad \forall n \in \mathbb{N}.$$

Then $\sum a_n b_n$ *is convergent.*

Definition 26. *A series of the form*

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n$$

or

$$\sum_{n=1}^{\infty} (-1)^{n} a_n,$$

where $a_n \geq 0$ *for all* $n \in \mathbb{N}$, *is called an* alternating series.

Example 59. *The following are alternating series:*

1. $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots.$

2. $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n}}{n^2}=-\frac{1}{1}+\frac{1}{4}-\frac{1}{9}+\frac{1}{16}-\cdots.$

3. $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2^n}=\frac{1}{2}-\frac{1}{4}+\frac{1}{8}-\frac{1}{16}+\cdots.$

Theorem 27 (Leibniz Test)**.** *Let (a_n) be a decreasing sequence of non-negative real numbers such that $a_n \to 0$. Then the alternating series*

$$\sum_{n=1}^{\infty}(-1)^{n-1}a_n$$

converges. Moreover, if

$$S=\sum_{n=1}^{\infty}(-1)^{n-1}a_n,$$

then

$$a_1-a_2 \leq S \leq a_1.$$

Proof. Let

$$s_n=\sum_{k=1}^{n}(-1)^{k-1}a_k.$$

Observe that

$$s_{2n}=(a_1-a_2)+\cdots+(a_{2n-1}-a_{2n}).$$

Since $a_{2n+1} \geq a_{2n+2}$,

$$s_{2n+2}-s_{2n}=a_{2n+1}-a_{2n+2} \geq 0.$$

Hence (s_{2n}) is increasing. Moreover,

$$s_{2n}=a_1-(a_2-a_3)-\cdots-(a_{2n-2}-a_{2n-1})-a_{2n} \leq a_1.$$

Thus (s_{2n}) is bounded above, and therefore converges. Let

$$s_{2n} \longrightarrow S.$$

Now,

$$s_{2n+1}=s_{2n}+a_{2n+1}.$$

Since $a_{2n+1} \to 0$, we obtain

$$s_{2n+1} \longrightarrow S.$$

Hence, both the even and odd subsequences of (s_n) converge to the same limit. Therefore,

$$s_n \longrightarrow S,$$

which proves that the series converges.
Finally,

$$a_1 - a_2 = s_2 \leq s_{2n} \leq S \leq s_{2n+1} \leq s_1 = a_1,$$

and letting $n \to \infty$ gives

$$a_1 - a_2 \leq S \leq a_1.$$

$\square$

Example 60. *Consider the alternating harmonic series*

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}.$$

Since the sequence $a_n = \dfrac{1}{n}$ *is decreasing and satisfies* $a_n \to 0$*, the Leibniz Test* (27) *implies that the series converges.*
However, the series is not *absolutely convergent, because*

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n},$$

which is the harmonic series and is divergent (see Example 18).
Thus, the alternating harmonic series is conditionally convergent.

Example 61. *Show that*

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-1/n}$$

is convergent.
Solution. *Let* $a_n = \frac{(-1)^{n+1}}{n}$ *and* $b_n = e^{-1/n}$*. The series* $\sum a_n$ *is convergent by the Leibniz Test* (27)*.*
Moreover, (b_n) *is increasing and*

$$0 < b_n < 1, \qquad \forall n \in \mathbb{N}.$$

Thus (b_n) *is bounded and monotone. Hence, by Abel's Test* (17)*,*

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-1/n}$$

is convergent.